CYBERSECURITY LAB SETUP

VirtualBox & Kali Linux — Week 1 Project

Intern Name	Rishu Satish Singh
Batch Code	B084H
Task ID	WK1-PM1
Organization	Networkwalks
Mentor	Waqas Karim, CCIE

Project Overview

This report documents the configuration of a Kali Linux virtual machine in Oracle VirtualBox for a cybersecurity practice environment. The VM is attached to a dedicated NAT Network named `NatNetwork`. The details below are based on the provided setup screenshots.

Purpose

The objective is to prepare a virtual environment for learning Linux, virtual networking, and future authorized cybersecurity exercises. The report records the VM resources and network settings visible in the provided images.

Lab Specifications

Component	Configuration
Virtualization Platform	Oracle VirtualBox
Guest OS	Kali Linux 2026.3
System Type	Debian (64-bit)
Allocated RAM	2048 MB (2 GB)
Processors	2
Video Memory	128 MB
Virtual Disk	VDI, 80.09 GB
Network Attachment	NAT Network
Network Name	NatNetwork
Network Adapter	Intel PRO/1000 MT Desktop (82540EM)
Promiscuous Mode	Allow All

Network Configuration

The provided IPv4 settings screen shows the connection name `Wired connection 1`, with the method set to Automatic (DHCP). The additional static-address table lists `10.0.0.2`, prefix `24`, and gateway `10.0.0.1`. The additional DNS field is blank in that screen. These are recorded as displayed settings and do not, by themselves, confirm the active address.

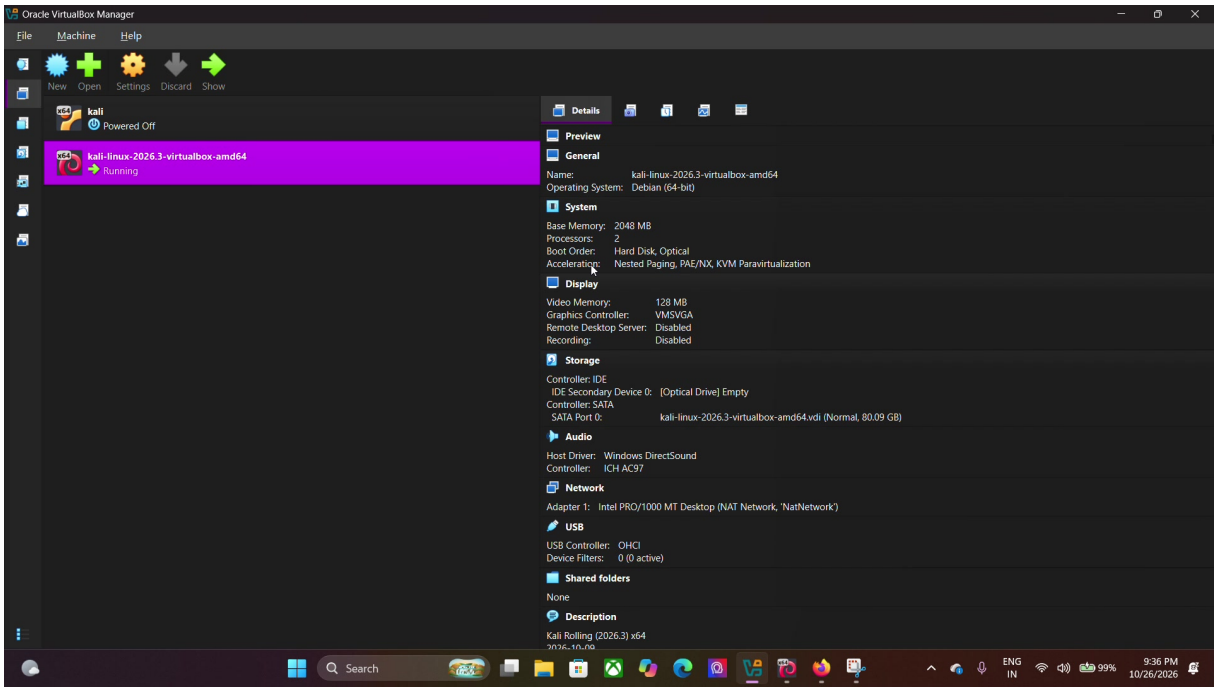
Network Adapter Settings

The VirtualBox adapter is attached to `NAT Network` and uses `NatNetwork`. The adapter model is Intel PRO/1000 MT Desktop (82540EM), Promiscuous Mode is set to Allow All, and Virtual Cable Connected is enabled in the displayed settings.

Setup Screenshots

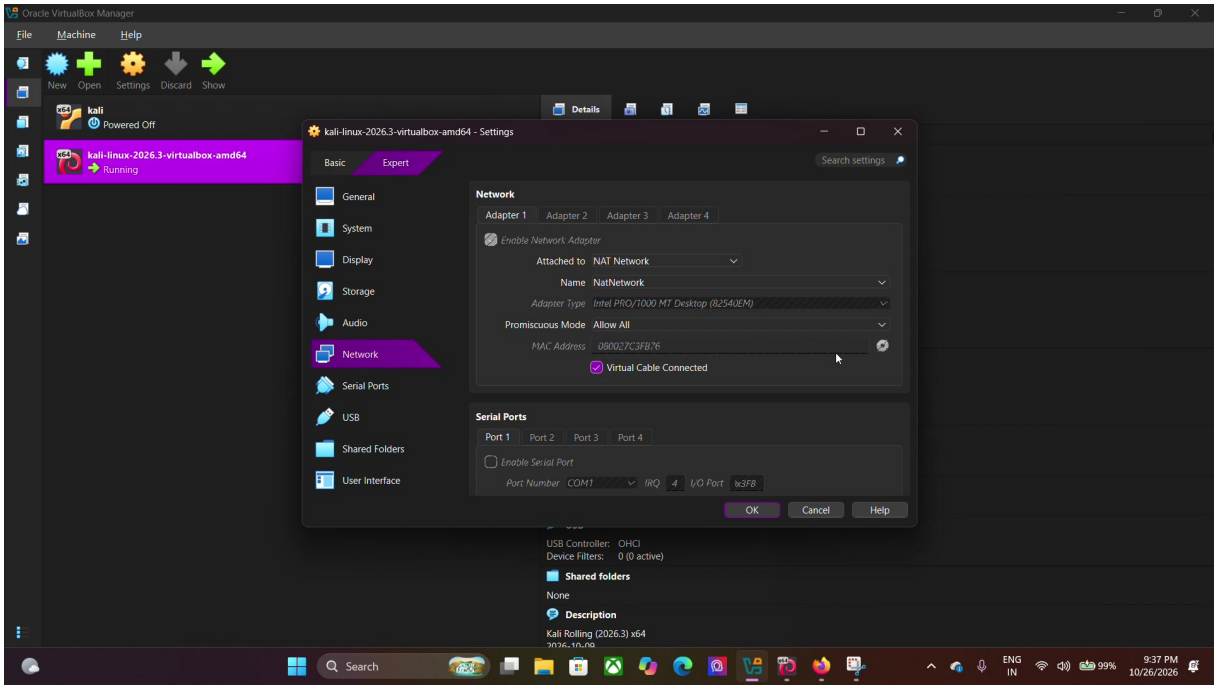
The following images were supplied with the project and document the configuration.

Figure 1. VirtualBox VM overview



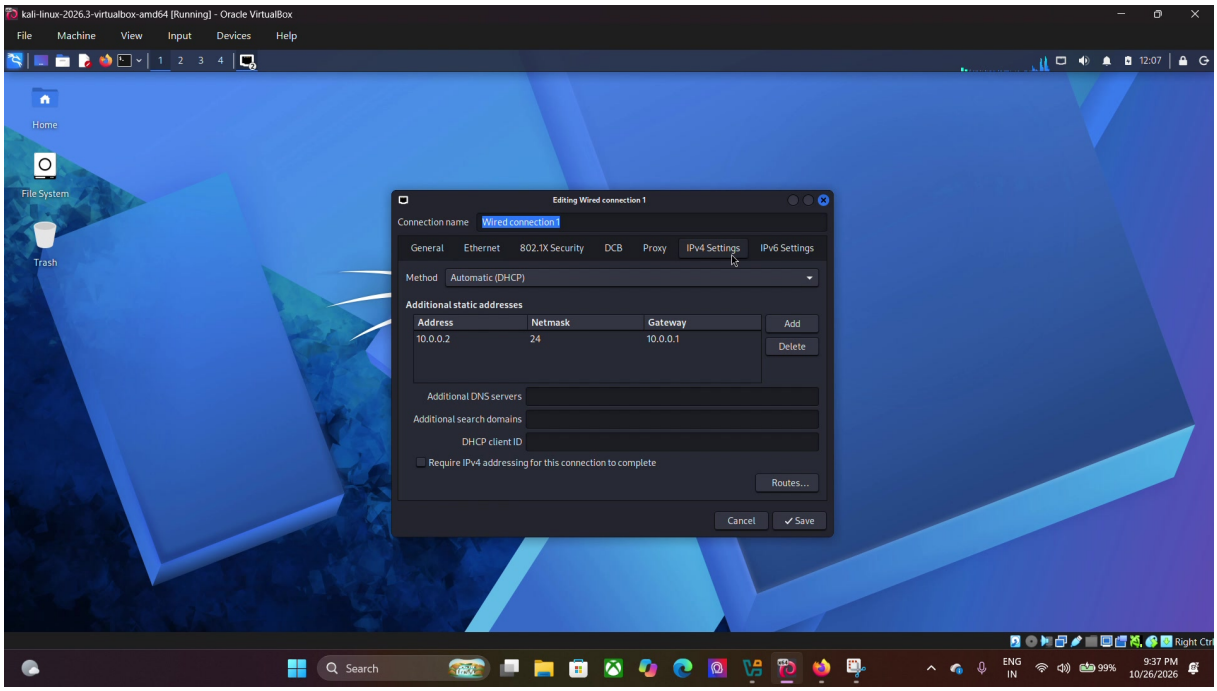
The overview displays Kali Linux 2026.3, Debian (64-bit), 2048 MB RAM, two processors, 128 MB video memory, a VDI disk of 80.09 GB, and the NAT Network attachment.

Figure 2. VirtualBox network adapter



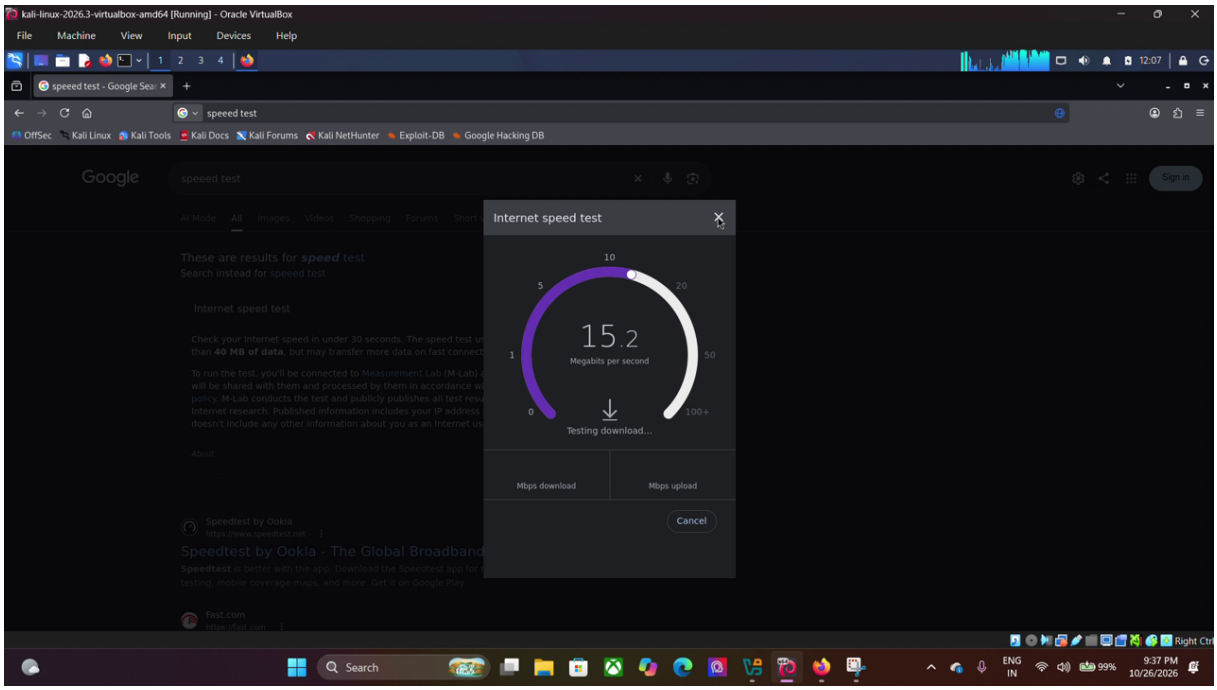
The adapter settings show NAT Network attachment, the network name NatNetwork, Intel PRO/1000 MT Desktop (82540EM), Promiscuous Mode Allow All, and Virtual Cable Connected enabled.

Figure 3. Kali Linux IPv4 settings



The connection is set to Automatic (DHCP), with an additional address 10.0.0.2/24 and gateway 10.0.0.1 listed.

Figure 4. Additional supplied screenshot



Included as provided for project documentation.

Learning Outcomes

This setup provided practical exposure to configuring a Kali Linux guest system, allocating virtual hardware resources, selecting a VirtualBox NAT Network, and reviewing IPv4 connection settings. The documented configuration serves as a starting point for future networking and cybersecurity labs.

Security and Ethical Use

This environment is intended for educational practice and authorized security testing only. Testing should be performed only on systems owned by the learner or where explicit permission has been granted.

Tools and Resources

Oracle VirtualBox: https://www.virtualbox.org/wiki/Downloads
Kali Linux: https://www.kali.org/get-kali/
7-Zip: https://www.7-zip.org/

Project Details

Field	Details
Program	Cybersecurity at Networkwalks
Week	01
Task ID	WK1-PM1
Project	Cybersecurity Lab Setup — VirtualBox & Kali Linux
Intern	Rishu Satish Singh
Batch	B084H
Mentor	Waqas Karim, CCIE